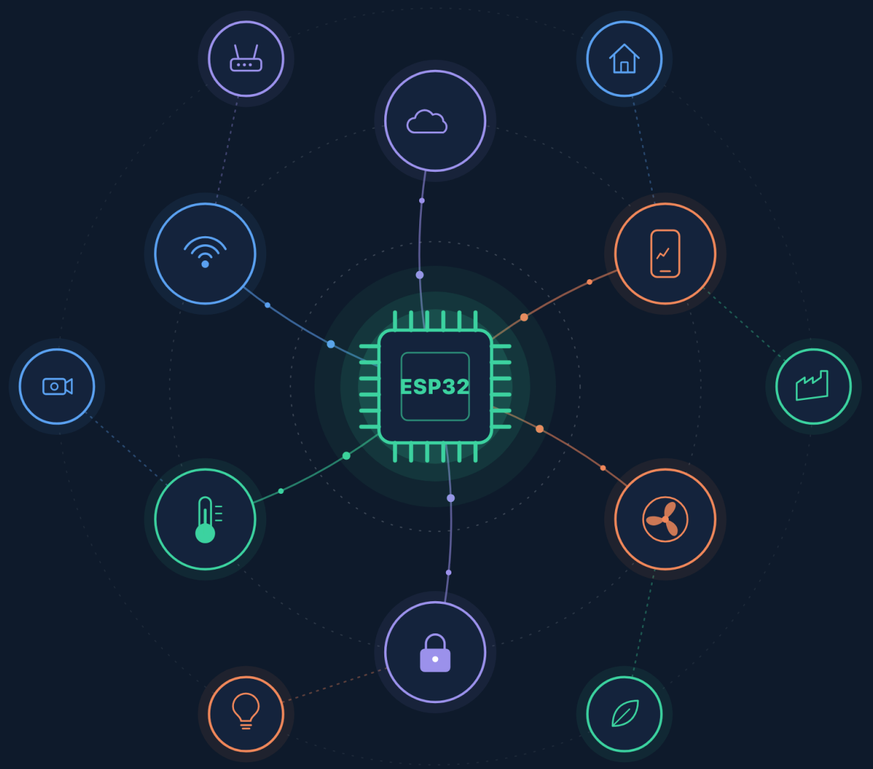
MODULE 6 · 45 MINUTES

Data Processing and Automation

How a system turns raw readings into decisions, and then acts on them by itself.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
Lecture

02
How it's built
Lecture

03
Sensors and actuators
Lab 2

04
Getting online
Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05
Cloud and dashboards
Lab 4

06
Automation
You are here

07
Security
Group activity

08
Real world + pitch
Lab 7

BY THE END OF THIS MODULE

You will be able to...

01 **Describe** how raw readings become a decision and an action

02 **Write** simple IF-THEN rules for a tower site

03 **Explain** hysteresis, and why edge rules work offline

04 **Decide** what runs automatically and what needs a person

05 **Build** a site monitor that switches on cooling by itself

THE BIG PICTURE

From reading to action



Raw data

31.4, 31.6, 31.5 °C in the shelter



Process

Check, clean and average



Rule

Is it above 28 °C?



Act or wait

Yes: cooling on + alert. No: keep watching.

Module 1’s loop: **sense** → **send** → **decide** → **act**. Today is **decide** and **act**.

AUTOMATION RULES: IF THIS, THEN THAT

Four automations for a tower site



Hot shelter

IF above 28 °C, **THEN** extra cooling + an alert.



Battery low

IF the battery is low, **THEN** start the generator.



Fuel theft?

IF fuel drops, generator off, **THEN** alert the NOC.



Service early

IF battery health falls, **THEN** book a service.

Energy is **15–20%** of a mobile operator’s operating costs (GSMA Intelligence, 2025).

PREDICTIVE MAINTENANCE

Service it before it fails

Battery health

The trend falls week by week: replace it in time.

Run-hours

The generator reaches its service hours: book a visit.

Cooling units

It works harder for the same temperature: check it.

Not on a fixed calendar, not after it breaks: **when the data says so**.

THE SAME IDEA IN OUR DATA CENTRES

Power failures cause the worst outages

No. 1

cause of serious outages: power

1 in 5

operators: last major outage cost over US$1 million

UPS systems, transfer switches and generators fail most often. Watch their health and fix them early.

Source: Uptime Institute, Annual Outage Analysis 2026

A FAMOUS EXAMPLE

DeepMind: less cooling energy from old data

Up to 40%

less energy used for cooling

1,000s

of sensors the sites already had

Machine learning on data the sensors **already collected**: temperatures, power, pump speeds.

Source: Google DeepMind blog, July 2016

Automated network fixes: the same pattern



Network data

Alarms, traffic and faults from the network itself.



Decide

Software spots a problem or a pattern.



Act

It changes network settings, or calls a person.

This is **network automation**, not IoT: no new devices. But the pattern is the same: **decide → act**.

WHY CLEANING MATTERS

One bad reading can cause a false alarm



Validate

A loose wire sends **85 °C**. Rule: ignore a jump of more than 10 °C.



Smooth

Near the limit: 27.9, 28.1, 27.8, 28.2. Average a few first.

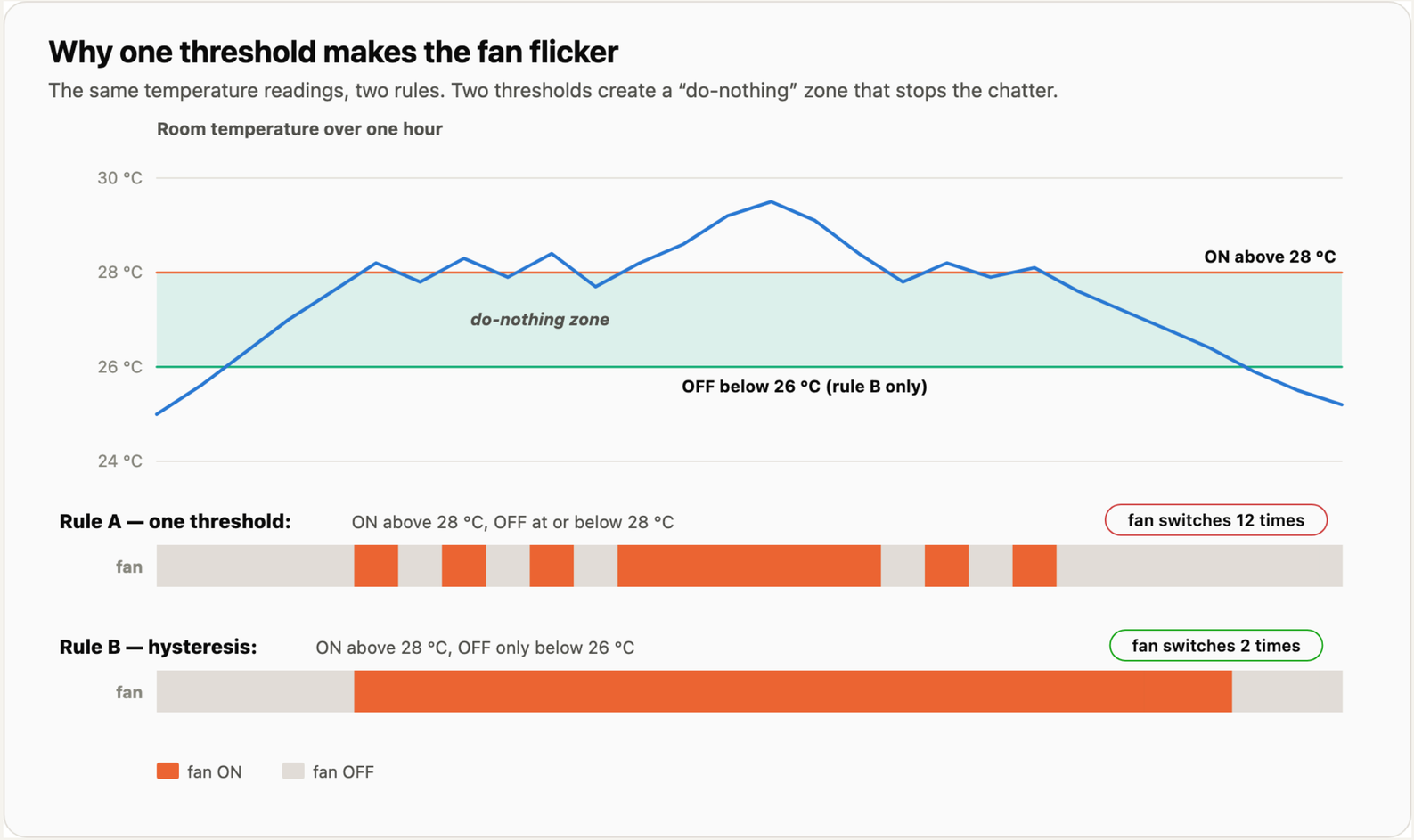
PAUSE AND CHECK

Your shelter sensor reports 24.1, 24.3, 24.2, 97.0, 24.2 °C. Which step should deal with the 97.0?

Validation or smoothing? Think on your own for 20 seconds.

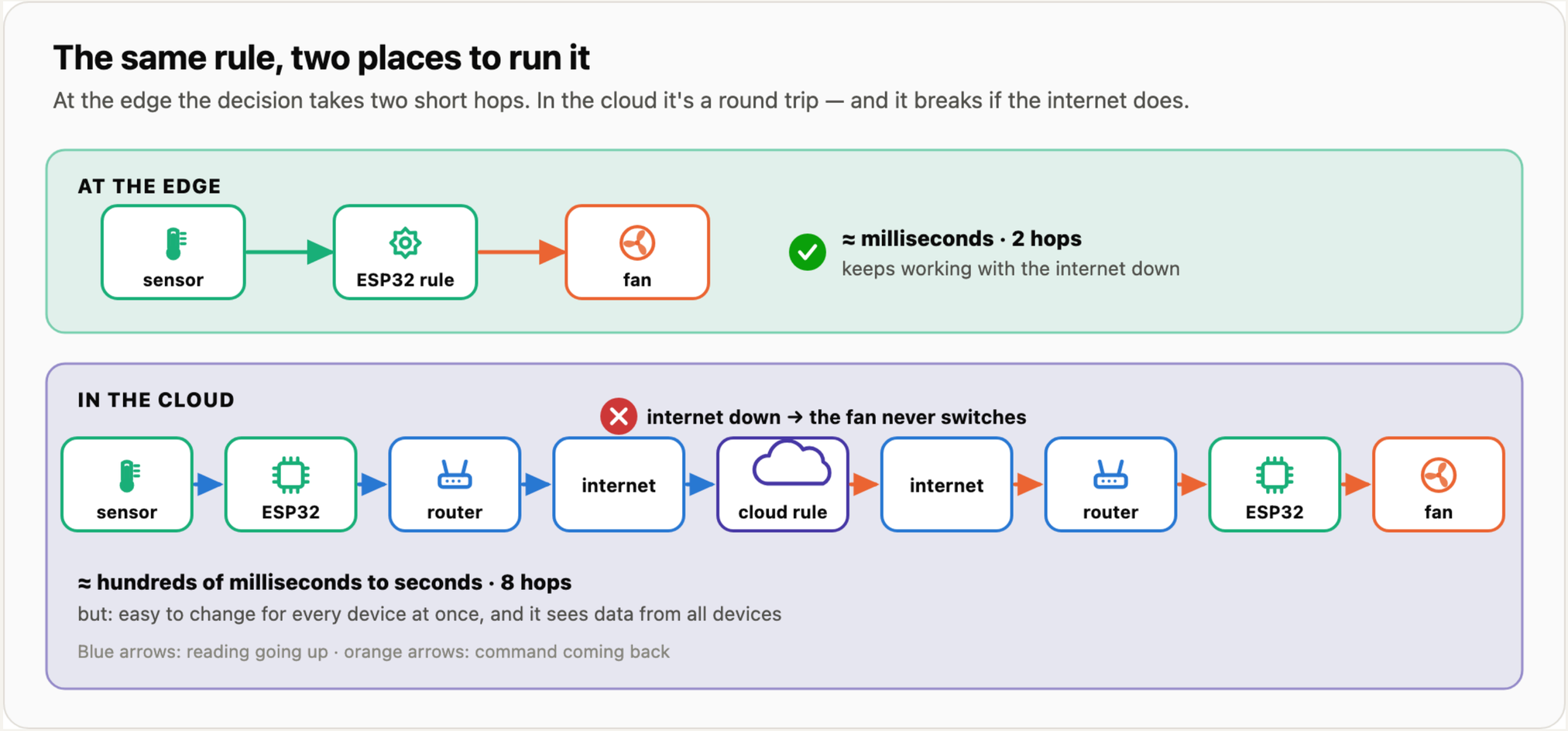
A CLASSIC RULE PROBLEM

Flickering, and the fix: hysteresis



EDGE VS CLOUD

The same rule, two places to run it



THE BUSINESS LENS

What a false alarm costs

Wasted trips

A long drive to a tower, for nothing.

Lost sleep

The on-call engineer is woken at 3 am.

Lost trust

Too many alarms, and people ignore the real one.

Wrong action

Good equipment is switched off.

Fewer false alarms: **clean** the data, use **hysteresis**, add a **time delay**.

DECIDE → ACT

Automatic, or a person confirms?

Situation	Action	Who acts?
Shelter above 28 °C	Switch on extra cooling	**Automatic**
Battery low, no mains	Start the generator	**Automatic**
Battery health falling	Book a service visit	**Person confirms**
Fuel drops, generator off	Send a team to the site	**Person confirms**
Data hall: three racks hot	Move or shut down servers	**Person confirms**

Automatic when it is **fast, safe and easy to undo**. A person confirms when it is **costly, risky or hard to undo**.

IN TELCO OPERATIONS

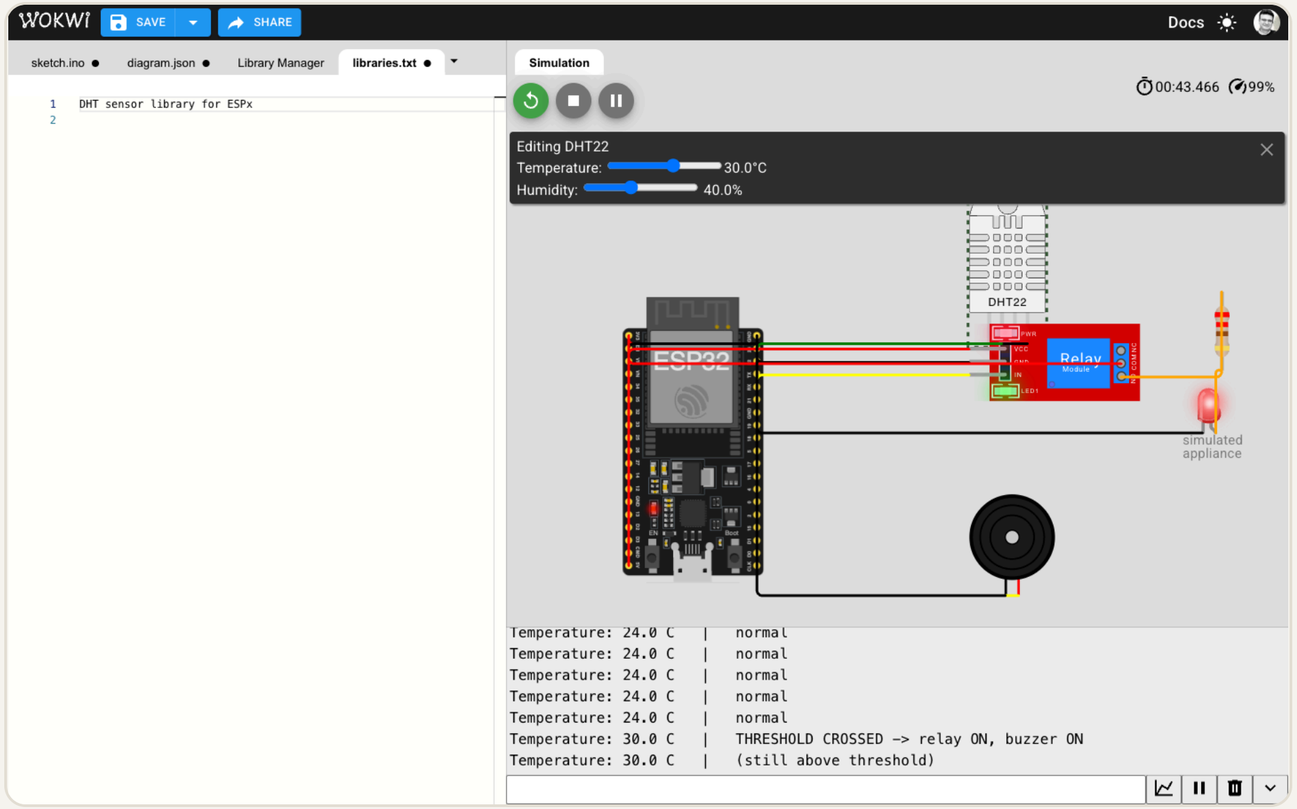
What automation changes at our sites

Today	With IoT	The benefit
Hot shelter: an engineer drives out	Cooling on by itself; NOC told	Faster fix, fewer site visits
Diesel: the generator runs too long	Runs only when the battery needs it	Less diesel, less wear
Maintenance: by calendar, or after failure	Service when the data says so	Fewer outages, better use of engineers

Canvas **box 4**: which row is closest to your team’s problem?

LAB 5 · 22 MINUTES HANDS-ON

Hot shelter: automatic cooling and alarm



1. Paste sketch.ino, diagram.json and libraries.txt
2. Press **Play**: a reading every 2 s, all “normal”
3. Drag the DHT22 above 28 °C: the shelter is hot
4. Relay = extra cooling **ON**; the buzzer alerts
5. Fill in canvas **box 4**: the decision and the action

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- A rule turns data into action: **IF** condition **THEN** action
- **Clean** the data first: one bad reading can cause a false alarm
- **Hysteresis** (ON at 28 °C, OFF at 26 °C) stops flickering
- **Edge** rules work offline; the **cloud** sees every site
- Site rules pay: cooling, generator, fuel theft, early service
- Automate what is safe and easy to undo; a **person confirms** the rest

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 Why does the shelter cooling switch ON at 28 °C but OFF only at 26 °C?

Q2 The link to a tower site is down. Where should the cooling rule run?

Q3 Fuel drops overnight while the generator was off. What should happen?

Q4 Which two readings tell you to service a site before it fails?

UP NEXT · MODULE 7

IoT Security and Best Practices

What can go wrong: how IoT devices get attacked, and how to protect tower sites, data centres and the network.

Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop